about its reproductive success and that of its relatives, rather than, for instance, its own longevity.

How shall we resolve this paradox of the two ways of looking at life? My own attempt to do so is spelled out in *The Extended Phenotype*, the book that, more than anything else I have achieved in my professional life, is my pride and joy. This chapter is a brief distillation of a few of the themes in that book, but really I'd almost rather you stopped reading now and switched to *The Extended Phenotype!*

On any sensible view of the matter Darwinian selection does not work on genes directly. DNA is cocooned in protein, swaddled in membranes, shielded from the world, and invisible to natural selection. If selection tried to choose DNA molecules directly it would hardly find any criterion by which to do so. All genes look alike, just as all recording tapes look alike. The important differences between genes emerge only in their *effects*. This usually means effects on the processes of embryonic development and hence on bodily form and behaviour. Successful genes are genes that, in the environment influenced by all the other genes in a shared embryo, have beneficial effects on that embryo. Beneficial means that they make the embryo likely to develop into a successful adult, an adult likely to reproduce and pass those very same genes on to future generations. The technical word *phenotype* is used for the bodily manifestation of a gene, the effect that a gene, in comparison with its alleles, has on the body, via development. The phenotypic effect of some particular gene might be, say, green eye colour. In practice most genes have more than one phenotypic effect, say green eye colour and curly hair. Natural selection favours some genes rather than others not because of the nature of the genes themselves, but because of their consequences —their phenotypic effects.

Darwinians have usually chosen to discuss genes whose phenotypic effects benefit, or penalize, the survival and reproduction of whole bodies. They have tended not to consider benefits to the gene itself. This is partly why the paradox at the heart of the theory doesn't normally make itself felt. For instance a gene may be successful through improving the running speed of a predator. The whole predator's body, including all its genes, is more successful because it runs faster. Its speed helps it survive to have children; and therfore more copies of all its genes, including the gene for fast running, are passed on. Here the